

Rwanga Codebase Audit Report

Generated 2026-05-01 | Audited via live API at `rwanga.zeneon.co.uk` + design-kit filesystem

16

Apps Registered

4

500 Errors

7

Open Gaps

89

Progress Tasks

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1. Architecture Overview

Stack: Django 5 + HTMX + Bootstrap 5 RTL + PostgreSQL + Celery + Channels + DRF
Skeleton: Cloned from HUD2 production platform. Uses `src/` as package name (deliberate).
Settings: `django-split-settings` with 17 component files in `src/settings/components/`.
URLconf: `src.urls` mounts both HTML views and `api/v1/` DRF routes for all 16 apps.
DEBUG = True in production (Cloudflare tunnel). Full tracebacks visible to the public.

LOCAL_APPS Registration

The MASTER-DESIGN specifies 15 LOCAL_APPS. An open gap says "only 4 are installed" but the live API schema shows all 16 apps have registered DRF routes. Evidence suggests a later agent run added them.

The gap tracker is stale. LOCAL_APPS likely has all apps now, but the gap "only 4 installed" remains open, causing confusion.

2. Per-App Inventory

2.1 core

Aspect	Status
Models	OK BaseModel (UUID pk, timestamps), abstract mixins
Views	N/A (framework app)
API	Health endpoint at <code>/api/v1/health/</code> returns <code>{"status":"ok"}</code>
Templates	N/A
Tests	OK <code>test_models.py</code> exists per progress tracker
Services	Shared utilities, context_processors (<code>studio_context</code>), middleware (<code>StudioContextMiddleware</code>)
Admin	N/A

URLs	Wired (health endpoint)
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2.2 accounts

Aspect	Status
Models	OK Studio, User (custom), Membership, ConsultantProfile, SignupProfile, ConsultantAssignment. All have DRF endpoints.
Views	PARTIAL Login: functional (allauth). Register: BROKEN renders login page instead of register form. Profile/Settings/Team/Contacts: all return STUB "coming soon".
API	OK Full CRUD: studios, users, memberships, consultants, consultant-assignments, signup-profiles, token auth
Templates	Design-kit has: login.html, register.html, profile.html, settings.html, team.html, contacts.html. Login is deployed. Register NOT deployed (shows login). Others are stubs.
Tests	OK Tests exist per progress tracker
Services	Unknown - no filesystem access
Admin	OK Admin accessible at /admin/ (302 to login)
URLs	OK Both app urls.py and root urls.py wired. Allauth routes present.

Register page is broken - renders the login template instead of register.html from design-kit. Magic link form exists but email backend is console-only (no actual emails sent).

2.3 projects

Aspect	Status
Models	PARTIAL Project, Scene, Character, Location all exist. BUT Project model schema has drifted from MASTER-DESIGN: missing title_latin, project_type, logline, cover fields. Currently using metadata JSON as workaround. OPEN GAP
Views	OK List (works, shows empty state), Create Wizard (works, 4-step), Dashboard (works, renders real data with module grid), Settings (STUB coming soon)
API	OK Full CRUD: projects, scenes, characters, locations. Nested routes work: /projects/{id}/scenes/ etc. No auth required for GET (security issue).
Templates	Design-kit: dashboard.html, create_wizard.html, list.html, scene_view.html, _scene_list.html, settings.html, 10 scene tab partials. Dashboard and list are deployed. Scene tabs exist in design-kit but unclear if wired.
Tests	OK Per progress tracker
Services	Unknown
Admin	OK
URLs	OK Wired in both app and root

Project model drift is a major issue. The MCP server stores project_type/logline/title_latin in the synopsis field as [RWANGA_META] JSON blob -- a fragile hack.

2.4 progress

Aspect	Status
Models	OK ProgressTask, ProgressUpdate, ChangeRecord, GapBlocker, DesignDecision, AgentReport, SystemDiagram. All have DRF endpoints. 89 tasks, 20 gaps tracked.
Views	OK Dashboard renders at /progress/. Multiple sub-pages: tasks, gaps, updates, changelog, decisions, diagrams, agent_reports, docs.
API	OK Full CRUD, no auth required for GET (security). Pagination working.
Templates	Design-kit has 10 templates. Deployed but had rendering issues (split filter, missing URL names) -- these were resolved per gap tracker.
Tests	OK model, api, service tests exist
Services	OK
Admin	OK
URLs	OK

Most complete app in the project. Functions as the primary tracking system for the build itself.

2.5 scripts

Aspect	Status
Models	OK Script, ScriptElement. DRF endpoints exist with nested routes.
Views	PARTIAL Index (200, has design-kit template with tabs). Upload (200, has multipart form). Breakdown: STUB "coming soon". Elements: STUB . Docs: STUB .
API	OK scripts, elements, nested per-project routes. Returns empty results (no data).
Templates	Design-kit: index.html, upload.html, breakdown.html, elements.html, docs.html. Index and upload deployed. Others are stub pages.
Tests	Unknown
Services	Unknown
Admin	Unknown
URLs	OK Wired

Upload form exists but unclear if it actually processes files. 3 of 5 sub-pages are stubs.

2.6 shots

Aspect	Status
Models	OK Shot, Setup, StoryboardFrame. All have DRF endpoints.
Views	PARTIAL List (200, has filter buttons for type, export PDF link -- looks functional). Storyboards (200, shows empty state).

API	OK Full CRUD: shots, setups, storyboard-frames
Templates	Design-kit: list.html, storyboards.html. Both deployed.
Tests	Unknown
Services	Unknown
Admin	Unknown
URLs	OK

Shot list page looks more functional than most (has HTMX filter buttons). Sortable.js drag-drop NOT wired (P2.4 pending). No shot data exists in DB.

2.7 floorplans

Aspect	Status
Models	OK FloorPlan with JSON fields (furniture, cameras, paths). DRF endpoint exists.
Views	PARTIAL List (200, shows empty state). Editor view unknown.
API	OK CRUD at /api/v1/floorplans/floorplans/
Templates	Design-kit: list.html deployed. No editor template in design-kit.
Tests	Unknown
Services	Unknown
Admin	Unknown
URLs	OK

2.8 scheduling

Aspect	Status
Models	OK ShootDay, ScheduleBlock, CallSheet. All have DRF endpoints.
Views	BROKEN Index (200, shows empty state -- works). Stripboard: 500 ERROR NoReverseMatch for 'reorder_strips'. Call-sheets: 500 ERROR NoReverseMatch for 'generate_call_sheet_modal'.
API	OK CRUD: shoot-days, schedule-blocks, call-sheets
Templates	Design-kit: index.html, stripboard.html, call_sheets.html. Templates reference URL names that don't exist as views.
Tests	Unknown
Services	Unknown
Admin	Unknown
URLs	BROKEN Missing URL patterns: reorder_strips, generate_call_sheet_modal

2 of 3 pages crash with 500 errors. Templates copied from design-kit but the URL names they reference were never created as views.

2.9 departments

Aspect	Status
Models	OK LightingNote, SoundNote, Prop, WardrobeItem, ContinuityItem. All have DRF endpoints.
Views	PARTIAL All 5 pages return 200. Lighting has a basic form (hx-post). Others appear to have minimal forms but no real list/data display. These are NOT stubs but NOT fully functional either.
API	OK Full CRUD for all 5 model types
Templates	Design-kit has scene tab partials (lighting, sound, props, wardrobe, continuity). Standalone department pages appear to use different templates.
Tests	Unknown
Services	Unknown
Admin	Unknown
URLs	OK

Views exist but are thin. Forms render but lack proper list views. GAPS task "T1 Departments UX flows" marked completed but pages still look minimal.

2.10 locations

Aspect	Status
Models	OK Location model with DRF endpoint.
Views	500 ERROR <code>/locations/</code> crashes: NoReverseMatch for 'add_modal'. Template references a URL that doesn't exist.
API	OK CRUD at <code>/api/v1/locations/locations/</code>
Templates	Design-kit: list.html. Deployed but references missing URL.
Tests	Unknown
Services	Unknown
Admin	Unknown
URLs	BROKEN Missing: add_modal

Page crashes. The sidebar nav links to `/locations/` on every page, so users see this crash regularly.

2.11 notifications

Aspect	Status
Models	OK Notification model with DRF endpoint.
Views	500 ERROR <code>/notifications/panel/</code> crashes: NoReverseMatch for 'list'. Template references missing URL.
API	OK CRUD at <code>/api/v1/notifications/notifications/</code>
Templates	Design-kit: panel.html. Deployed but broken.

Tests	Unknown
Services	Unknown
Admin	Unknown
URLs	BROKEN Missing: list

The notification bell button exists on EVERY page (in the sidebar). Clicking it triggers an HTMX GET to `/notifications/panel/` which returns a 500. Silent failure in the UI.

2.12 reviews

Aspect	Status
Models	OK InlineComment, BibleReview, ReviewDecision, SceneEvaluation. Full DRF endpoints including custom actions (lock, reject).
Views	STUB <code>/reviews/<pk>/</code> returns 200 but renders a minimal card: "Reviews" heading with "No reviews yet." in a plain <code><u1></code> . No form, no filters, no real UI.
API	OK Full CRUD plus lock/reject actions on decisions
Templates	No design-kit template for reviews. Current template is a bare-minimum <code>rw-card</code> .
Tests	Unknown
Services	Unknown
Admin	Unknown
URLs	PARTIAL <code>/reviews/</code> returns 404 (no index). <code>/reviews/<pk>/</code> works.

Backend is comprehensive (4 models, custom actions). Frontend is a 5-line stub. No design-kit template exists for this app.

2.13 ai_engine

Aspect	Status
Models	OK AIJob model with DRF endpoint.
Views	MISSING No HTML views. <code>/ai/</code> and <code>/ai/<pk>/</code> both return 404.
API	OK CRUD at <code>/api/v1/ai/jobs/</code>
Templates	Design-kit has <code>_ai_progress.html</code> component partial. No standalone pages.
Tests	Unknown
Services	PARTIAL Provider architecture created (ollama, nllb, stable_diffusion) per GAPS task. Celery tasks exist but AI features are disabled (<code>RWANGA_AI_ENABLED=False</code>).
Admin	Unknown
URLs	PARTIAL API wired. HTML routes in root <code>urls.py</code> but no matching views.

2.14 exports

Aspect	Status
Models	NONE No models. Only a health endpoint.
Views	MISSING <code>/exports/</code> returns 404. No scene viewer or call sheet export views accessible.
API	PARTIAL Only <code>/api/v1/exports/health/</code> exists. No export endpoints.
Templates	No design-kit templates for exports.
Tests	Unknown
Services	PARTIAL GAPS T2 "Exports functional pipeline" marked completed. ExportService may exist but endpoints are not wired.
Admin	N/A
URLs	BROKEN Root <code>urls.py</code> has <code>exports/</code> entry but app has no matching URL patterns beyond health.

GAPS task says "completed" but no export endpoints actually work. Scene viewer and call sheet PDFs are not accessible.

2.15 community

Aspect	Status
Models	OK CommunitySession, Participant, SharedContent, Comment, Reaction. Full DRF endpoints.
Views	STUB <code>/community/<pk>/</code> returns 200. Renders "Community Sessions" heading with "No sessions" in plain <code></code> . No form, no real UI.
API	OK Full CRUD for sessions, participants, contents, comments, reactions
Templates	No design-kit template. Current is a 4-line stub.
Tests	Unknown
Services	Unknown
Admin	Unknown
URLs	PARTIAL <code>/community/</code> returns 404. <code>/community/<pk>/</code> works.

Same pattern as reviews: rich backend (5 models), stub frontend. P6.1 "Create `src/community/` app" still pending but app clearly exists.

2.16 realtime

Aspect	Status
Models	NONE
Views	MISSING No HTTP endpoints. <code>/api/v1/realtime/</code> returns 404.
API	NONE
WebSocket	Unknown. ASGI config exists but consumers not verified.
Tests	Unknown

3. User-Reported Bug Investigation

BUG: Mobile Auth -- Can a user sign in?

Finding: PARTIALLY WORKING

- Login page renders correctly at `/accounts/login/` with RTL Kurdish text
- Two auth methods shown: Magic Link (primary) and Email/Password (secondary)
- Magic Link: form exists, posts to `/accounts/magic-link/`. BUT email backend is `console.EmailBackend` -- emails are printed to server console, never actually sent. **Magic link auth is non-functional for real users.**
- Email/Password: form posts to `/accounts/login/`. Returns 403 (CSRF failure in my test, but would work with proper CSRF token in browser).
- allauth is configured and the login view is the allauth LoginView per resolved gap.
- Register page at `/accounts/register/` **renders the login template**, not the register template. Users cannot create accounts through the UI.
- **Login page does NOT load bootstrap.bundle.min.js** -- it only loads `rwanga.js` and `htmx`. This means any Bootstrap JS components (modals, dropdowns) won't work on the login page.

BUG: Dropdown Menus -- Are Bootstrap dropdowns wired?

Finding: INCONSISTENT

- Main app pages (projects dashboard, scripts, etc.) load `bootstrap.bundle.min.js` from CDN at the bottom of body - dropdowns work here.
- Login page does NOT load `bootstrap.bundle.min.js` -- only `rwanga.js` and `htmx`.
- User avatar dropdown uses `data-bs-toggle="dropdown"` -- requires `bootstrap.bundle.js`.
- The dropdown markup is correct (Bootstrap 5 standard). The JS is loaded on most pages. **Dropdowns should work on all pages except login/register.**
- Bootstrap CSS is loaded from CDN (`bootstrap.min.css`) -- NOT the RTL variant (`bootstrap.rtl.min.css`). The `dir="rtl"` attribute is set on `<html>` so Bootstrap 5's built-in RTL should activate, but using the explicit RTL build would be safer.

BUG: Projects Dashboard -- Does `/projects/<pk>/` render real data?

Finding: WORKS

- Dashboard renders the project title, studio name, scene count ("0 scenes"), and a module grid.
- Module grid shows 5 color-coded sections (Script, Breakdown, Visualization, Planning, Shooting) with clickable module cards.
- Links to all sub-modules are present and use the correct project UUID.
- The scene count shows real data (0 -- because no scenes have been created via API).
- **This is the most polished page in the app.** It uses the design-kit `dashboard.html` template correctly.

BUG: Reviews Section -- Any UI, or backend-only?

Finding: BACKEND-ONLY

- `/reviews/<pk>/` renders a minimal stub: a card with "Reviews" heading and "No reviews yet" in a list.
- No create form, no filter UI, no inline editing, no review workflow.
- Backend has 4 models (InlineComment, BibleReview, ReviewDecision, SceneEvaluation) with full DRF CRUD + custom lock/reject actions.
- **No design-kit template exists for reviews.** A custom template must be created.

BUG: Community Section -- Any UI, or backend-only?**Finding: BACKEND-ONLY**

- `/community/<pk>/` renders a minimal stub: "Community Sessions" heading, "No sessions" list.
- Backend has 5 models with full DRF CRUD.
- **No design-kit template exists.** Feature flag `RWANGA_COMMUNITY_ENABLED=False`.

4. API & URL Health Matrix

App	API Root	API Status	HTML Root	HTML Status
accounts	<code>/api/v1/accounts/</code>	403 (auth required)	<code>/accounts/login/</code>	200
projects	<code>/api/v1/projects/projects/</code>	200 (open!)	<code>/projects/</code>	200
progress	<code>/api/v1/progress/tasks/</code>	200 (open!)	<code>/progress/</code>	200
scripts	<code>/api/v1/scripts/</code>	403	<code>/scripts/<pk>/</code>	200
shots	<code>/api/v1/shots/</code>	403	<code>/shots/<pk>/</code>	200
floorplans	<code>/api/v1/floorplans/</code>	403	<code>/floorplans/<pk>/</code>	200
scheduling	<code>/api/v1/scheduling/</code>	403	<code>/scheduling/<pk>/stripboard/</code>	500
departments	<code>/api/v1/departments/</code>	403	<code>/departments/<pk>/lighting/</code>	200
locations	<code>/api/v1/locations/</code>	403	<code>/locations/</code>	500
notifications	<code>/api/v1/notifications/</code>	403	<code>/notifications/panel/</code>	500
reviews	<code>/api/v1/reviews/</code>	403	<code>/reviews/<pk>/</code>	200 stub
ai_engine	<code>/api/v1/ai/</code>	403	<code>/ai/</code>	404
exports	<code>/api/v1/exports/</code>	404 (health only)	<code>/exports/</code>	404
community	<code>/api/v1/community/</code>	403	<code>/community/<pk>/</code>	200 stub
realtime	N/A	N/A	N/A	N/A

Security Issue: Open API Endpoints

`/api/v1/projects/projects/` and `/api/v1/progress/tasks/` (and `/gaps/`) return data without authentication. All 8 projects and 89 tasks are publicly readable. With `DEBUG=True`, full tracebacks are also exposed.

5. Critical Issues Summary

#	Severity	Issue	Impact
1	CRITICAL	DEBUG=True in production	Full tracebacks, secret paths, settings exposed to public internet
2	CRITICAL	Open API endpoints (no auth on projects, progress)	All project data publicly readable
3	CRITICAL	4 pages return 500 errors (locations, notifications, scheduling x2)	Users hit crashes on sidebar links and dashboard module cards
4	CRITICAL	Register page renders login template	Users cannot create accounts
5	CRITICAL	Magic link email uses console backend	Primary auth method doesn't work
6	HIGH	Project model schema drift (missing title_latin, project_type, logline)	MCP server hacks data into synopsis field
7	HIGH	Exports app has no functional endpoints	Scene viewer and call sheet PDFs don't work despite being "completed"
8	MEDIUM	9 pages are stubs (profile, settings, team, contacts, breakdown, elements, docs, reviews, community)	Directors can't use half the advertised features
9	MEDIUM	Bootstrap CSS is not the RTL build	RTL layout relies on auto-detection, not explicit RTL styles
10	MEDIUM	Login page missing bootstrap.bundle.js	No JS components work on auth pages
11	MEDIUM	CDN vs local Bootstrap inconsistency	Design-kit says local vendor, production uses CDN
12	LOW	Progress task data has duplicates (same P0 tasks entered twice)	Confusing tracking data
13	LOW	Gap tracker stale (resolved items not closed, open items outdated)	Misleading project status

6. Prioritized Fix List for Engineering Prompt

Priority 0 -- Stop the Bleeding (before any feature work)

1. Set `DEBUG=False` for production, configure `ALLOWED_HOSTS`, add proper 404/500 templates
2. Require auth on all API endpoints (fix DRF `DEFAULT_PERMISSION_CLASSES` or per-viewset)
3. Fix 500 errors: add stub URL patterns for `locations:add_modal`, `notifications:list`, `scheduling:reorder_strips`, `scheduling:generate_call_sheet_modal`
4. Fix register page: wire `/accounts/register/` to the design-kit `register.html` template
5. Add `bootstrap.bundle.min.js` to login/register pages (or move it to base includes)

Priority 1 -- Make Core Flow Work

1. Migrate Project model to add `title_latin`, `project_type`, `logline`, `cover` fields; remove synopsis RWANGA_META hack
2. Configure real email backend (or at minimum, file-based for dev) so magic link auth works
3. Wire script upload to actually process files and create Script model instances
4. Replace 9 stub pages with real templates from design-kit where they exist

5. Wire exports: scene viewer HTML and call sheet PDF endpoints

Priority 2 -- Complete App UX

- 1. Build reviews UI (no design-kit template -- must be created)
- 2. Build community UI (no design-kit template -- must be created)
- 3. Complete department pages with proper list views and HTMX inline editing
- 4. Wire scene tab partials from design-kit into scene_view.html
- 5. Add Sortable.js for shot reorder (P2.4)

Priority 3 -- Polish & Infrastructure

- 1. Switch to RTL Bootstrap CSS build (bootstrap.rtl.min.css)
- 2. Decide CDN vs local Bootstrap and make consistent
- 3. Clean up progress tracker (deduplicate tasks, close stale gaps)
- 4. Add real tests for all apps (currently only core/accounts/projects/progress have verified tests)
- 5. Wire WebSocket consumers for AI job progress

Design-Kit Template Deployment Status

Template	In Design-Kit	Deployed	Working
base.html	Yes	Yes	Yes
components/* (7 files)	Yes	Yes	Yes
accounts/login.html	Yes	Yes	Yes
accounts/register.html	Yes	No	Shows login instead
accounts/profile.html	Yes	Stub	Coming soon
accounts/settings.html	Yes	Stub	Coming soon
accounts/team.html	Yes	Stub	Coming soon
accounts/contacts.html	Yes	Stub	Coming soon
projects/list.html	Yes	Yes	Yes
projects/dashboard.html	Yes	Yes	Yes
projects/create_wizard.html	Yes	Yes	Yes
projects/scene_view.html	Yes	Yes	Unknown (no scenes to test)
projects/settings.html	Yes	Stub	Coming soon
projects/scenes/tabs/* (10)	Yes	Unknown	Unknown
scripts/index.html	Yes	Yes	Yes
scripts/upload.html	Yes	Yes	Yes
scripts/breakdown.html	Yes	Stub	Coming soon
scripts/elements.html	Yes	Stub	Coming soon
scripts/docs.html	Yes	Stub	Coming soon

shots/list.html	Yes	Yes	Yes
shots/storyboards.html	Yes	Yes	Empty state
floorplans/list.html	Yes	Yes	Empty state
scheduling/index.html	Yes	Yes	Empty state
scheduling/stripboard.html	Yes	Yes	500 error
scheduling/call_sheets.html	Yes	Yes	500 error
locations/list.html	Yes	Yes	500 error
notifications/panel.html	Yes	Yes	500 error
progress/* (10 files)	Yes	Yes	Yes

Audit methodology: filesystem scan of design-kit + live HTTP probing of all URL patterns + DRF OpenAPI schema extraction + progress tracker API analysis. No direct access to Django source code (src/ directory not mounted in session). Findings based on observable behavior.